

A Spectral-Calculus Extension of the Mild Generalized Spin–Boson Theorem

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Status. `partial_result_likely_valid`. This packet proves a conservative extension of the source paper’s Theorem 1.3 to a separable infinite-dimensional spin space with bounded commuting normal interaction operators. It does not address the source paper’s Case 2 or Case 3 ultraviolet-divergence problems.

Source signal. The source paper is Lill–Lonigro [1], arXiv:2307.14727. On page 6, Remark 1.5 says that the finite-dimensional proof should extend to infinite-dimensional spin space when the interaction operators are bounded and satisfy Assumption 1.1, but that Proposition 4.4 has to be replaced by a spectral-calculus argument.

To see the latter statement, first observe that if some b_j has full rank, then $\text{Ker } b_j = \{0\} \Rightarrow \text{Ker } B_j = \{0\} \Rightarrow \bigcap_j \text{Ker } B_j = \{0\}$. Conversely, if no b_j has full rank, then we can find one $0 \neq u^{(j)} \in \mathbb{C}^d$ for each b_j , such that $b_j u^{(j)} = 0$. But then, $0 \neq u^{(1)} \otimes \dots \otimes u^{(N)} \in \bigcap_j \text{Ker } B_j = \{0\}$. $\triangle$

Remark 1.5 (Infinite-dimensional $\mathfrak{h}$). It is not too difficult to see that our result also holds true if $\mathfrak{h}$ is infinite-dimensional and B_j are compact operators satisfying Assumption 1.1. In fact, the properties of B_j only enter in Propositions 4.3 and 4.4, as well as in Eq. (74). In the latter equation, the boundedness of all B_j is exploited to prove that certain operators are closable². Proposition 4.3 also holds true for B_j being generic bounded operators. In Proposition 4.4, we construct a common eigenbasis of $\{B_j\}_{j=1}^N$, which would also be possible if B_j were compact operators. It is easy to see that in this case, the proof of Proposition 4.4 (with an infinite eigenbasis) would still go through.

In the case in which all B_j are bounded and satisfy Assumption 1.1, the proof of Proposition 4.4 has to be modified using spectral calculus in order to obtain an analogue of Theorem 1.3. This generalization is straightforward but technical. To keep the current article short, we postpone it to a future publication. $\triangle$

2. SOME EXAMPLES

Before starting with the proof of Theorem 1.3, let us discuss some particular instances of Hamiltonians in the form (6) to which Theorem 1.3 applies.

The σ_x spin-boson model. Consider the following expression on $\mathbb{C}^2 \otimes \mathcal{F}$:

$$H = \frac{\eta}{2} \sigma_z \otimes I_{\mathcal{F}} + I_{\mathbb{C}^2} \otimes d\Gamma(\omega) + \sigma_x \otimes (a^*(f) + a(f)), \quad (17)$$

where $f \in \mathcal{H}_{-s}$, $s \leq 1$, and $\eta \geq 0$; furthermore, σ_x and σ_z denote the first and third Pauli matrices, respectively:

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad (18)$$

This is the model usually referred to as “the” spin-boson model in the mathematical literature, cf. [2–9]. Physically, it describes a two-level system (qubit), whose energy levels differ by a quantity η , undergoing *decay* dynamics as a result of its interaction with the boson field; for example, it is employed in quantum optics to describe the interaction of a two-level atom with an electromagnetic field. In the monochromatic case ($\mathcal{H} = \mathbb{C}$), it reduces to the well-known Rabi model [73–76].

This model clearly belongs to the class of operators analyzed in the present work, corresponding to the case $D = 2$, $N = 1$, and with $K = \eta/2 \sigma_z$, $B = \sigma_x$ (thus $A = \sigma_x \otimes a(f)$). As such, Theorem 1.3 applies. Using the obvious isomorphism $\mathfrak{H} = \mathbb{C}^2 \otimes \mathcal{F} \simeq \mathcal{F} \oplus \mathcal{F}$ —that is, representing the most general element of $\mathfrak{H}$ as

$$\Psi = \begin{bmatrix} \Psi_e \\ \Psi_g \end{bmatrix} : \quad \Psi_e, \Psi_g \in \mathcal{F}, \quad (19)$$

with Ψ_e and Ψ_g respectively playing the role of the “excited” and “ground” wavefunction of the boson field, we can conveniently rewrite Eq. (17) as

$$\begin{aligned} H &= \begin{bmatrix} d\Gamma(\omega) + \frac{\eta}{2} & a^*(f) + a(f) \\ a^*(f) + a(f) & d\Gamma(\omega) - \frac{\eta}{2} \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} d\Gamma(\omega) + \frac{\eta}{2} & \\ & d\Gamma(\omega) - \frac{\eta}{2} \end{bmatrix}}_{H_{\text{free}}} + \underbrace{\begin{bmatrix} & a^*(f) \\ a^*(f) & \end{bmatrix}}_{A^*} + \underbrace{\begin{bmatrix} & a(f) \\ a(f) & \end{bmatrix}}_A. \end{aligned} \quad (20)$$

²Here, we are adopting the following nomenclature: An operator A on a Hilbert space $\mathfrak{H}$, with domain $\text{Dom}(A)$, is called closable if and only if there exists a closed operator $B : \text{Dom}(B) \subseteq \mathfrak{H} \rightarrow \mathfrak{H}$, $\text{Dom}(B) \supseteq \text{Dom}(A)$ that extends A (see e.g. [72, Definition 1.3]).

Problem Extracted. Give the omitted spectral-calculus proof of the infinite-dimensional spin-space analogue of Theorem 1.3 in the mild-divergence Case 1, at least under the same commuting-normal/common-kernel hypotheses on the interaction operators.

The Result

Let $\mathcal{H}$ be the one-particle Hilbert space, let $\omega \geq m_0 > 0$ be the dispersion relation, and let $\mathcal{F} = \mathcal{F}^+(\mathcal{H})$ be the bosonic Fock space. For $r \in \mathbb{R}$ write $\mathcal{H}_r$ and $\mathcal{F}_r$ for the Hilbert scales used in [1]. Let $\mathfrak{h}$ be a separable Hilbert space and let $K \in \mathcal{B}(\mathfrak{h})$ be self-adjoint. After adding a scalar constant, assume for notational simplicity that

$$H_{\text{free}} = K \otimes I_{\mathcal{F}} + I_{\mathfrak{h}} \otimes d\Gamma(\omega)$$

is nonnegative. This harmless shift does not change the domain formula below. Put $\mathfrak{H} = \mathfrak{h} \otimes \mathcal{F}$ and let $\mathfrak{H}_r$ be the Hilbert scale generated by H_{free} .

Theorem 1. Fix $0 < s \leq 1$. Let $f_1, \dots, f_N \in \mathcal{H}_{-s} \setminus \mathcal{H}$ be $\mathcal{H}$ -independent in the sense of Definition 1.2 of [1]. Let $B_1, \dots, B_N \in \mathcal{B}(\mathfrak{h})$ be bounded normal operators such that $B_j B_\ell = B_\ell B_j$ for all j, ℓ and

$$\bigcap_{j=1}^N \text{Ker } B_j = \{0\}.$$

Define

$$A = \sum_{j=1}^N B_j^* \otimes a(f_j), \quad A^* = \sum_{j=1}^N B_j \otimes a^*(f_j),$$

where the singular creation and annihilation operators are interpreted as in [1, Proposition 4.1]. Then the formal Hamiltonian

$$H = H_{\text{free}} + A^* + A$$

has the same conclusions as [1, Theorem 1.3]:

$$\text{Dom}(H) = \left\{ \Psi \in \mathfrak{H}_{2-s} : \left[1 + (H_{\text{free}} - z_0)^{-1} A^* \right] \Psi \in \text{Dom}(H_{\text{free}}) \right\},$$

for any real $z_0 \in \rho(H_{\text{free}})$, and H is self-adjoint on this domain. Moreover, for each j there are $f_{j,n} \in \mathcal{H}$ with $\|f_{j,n} - f_j\|_{\mathcal{H}_{-s}} \rightarrow 0$ such that, with

$$A_n = \sum_{j=1}^N B_j^* \otimes a(f_{j,n}), \quad A_n^* = \sum_{j=1}^N B_j \otimes a^*(f_{j,n}),$$

the cutoff Hamiltonians $H_{\text{free}} + A_n^* + A_n$ converge to H in norm resolvent sense.

Proof Intuition

The finite-dimensional source proof uses the simultaneous diagonalization of the matrices B_j . On each common eigenspace, A reduces to one annihilation operator $a(\sum b_j f_j)$; the source's Lemma 4.2 then gives dense range, because $\mathcal{H}$ -independence keeps this linear combination outside $\mathcal{H}$ whenever the coefficient vector is nonzero.

For bounded commuting normal operators, the joint spectral theorem gives the same picture with direct integrals instead of a finite direct sum. The common kernel condition says that the joint spectral point $(0, \dots, 0)$ has zero spectral projection. Thus almost every fiber sees a nonzero coefficient vector, and the same annihilation-operator range lemma works fiber by fiber. Orthogonal complements then glue the fiberwise density into global density.

Proof

We only spell out the points in [1] where finite dimensionality entered. All references to equations and propositions in this proof are to the source paper unless otherwise stated.

Boundedness and the kernel condition

Since K is bounded self-adjoint, the scale $\mathfrak{H}_r$ is equivalent to the tensor scale $\mathfrak{h} \otimes \mathcal{F}_r$ for the values r used below. Proposition 4.1 of [1] gives $a(f_j) \in \mathcal{B}(\mathcal{F}_s, \mathcal{F})$ and $a^*(f_j) \in \mathcal{B}(\mathcal{F}, \mathcal{F}_{-s})$. Because there are only finitely many bounded operators B_j , we have

$$A \in \mathcal{B}(\mathfrak{H}_s, \mathfrak{H}), \quad A^* \in \mathcal{B}(\mathfrak{H}, \mathfrak{H}_{-s}).$$

The same coherent-vector proof as Proposition 4.3 of [1] gives density of $\text{Ker}(A|_{\mathfrak{H}_2})$. Indeed, if

$$\mathcal{D}_0 = \{g \in \mathcal{H}_2 : \langle f_j, g \rangle = 0 \text{ for all } j\},$$

then $\mathcal{D}_0$ is dense in $\mathcal{H}$ by $\mathcal{H}$ -independence, and $\mathfrak{h} \otimes \mathcal{E}(\mathcal{D}_0)$ is dense in $\mathfrak{H}$ and contained in $\text{Ker}(A|_{\mathfrak{H}_2})$.

Dense range via the joint spectral theorem

Let E be the joint spectral measure of the commuting normal tuple $(B_1, \dots, B_N)$ on its compact joint spectrum $\Sigma \subset \mathbb{C}^N$. Choose a scalar spectral measure ν and a direct-integral representation

$$\mathfrak{h} \simeq \int_{\Sigma}^{\oplus} \mathfrak{h}_{\lambda} d\nu(\lambda)$$

in which B_j acts as multiplication by the coordinate λ_j . The common kernel hypothesis implies

$$E(\{(0, \dots, 0)\}) = 0,$$

because this projection is the kernel projection of $\sum_j B_j^* B_j$.

Tensoring with $\mathcal{F}$ gives

$$\mathfrak{H} \simeq \int_{\Sigma}^{\oplus} (\mathfrak{h}_{\lambda} \otimes \mathcal{F}) d\nu(\lambda).$$

On the dense decomposable domain with Fock component in $\mathcal{F}_2$, the operator A is decomposable. Its fiber at λ is

$$A(\lambda) = I_{\mathfrak{h}_{\lambda}} \otimes a(g_{\lambda}),$$

where g_{λ} is the corresponding nonzero linear combination of $f_1, \dots, f_N$ with coefficient vector λ (up to the harmless conjugation dictated by the inner-product convention). Since $\lambda \neq 0$ for ν -almost every λ and the family $f_1, \dots, f_N$ is $\mathcal{H}$ -independent, we have

$$g_{\lambda} \in \mathcal{H}_{-s} \setminus \mathcal{H}, \quad g_{\lambda} \neq 0$$

for almost every λ .

By Lemma 4.2(ii) of [1], $\text{Ran}(a(g_\lambda)|_{\mathcal{F}_2})$ is dense in $\mathcal{F}$ for almost every λ . Hence

$$\text{Ran}\left(I_{\mathfrak{h}_\lambda} \otimes a(g_\lambda) \big|_{\mathfrak{h}_\lambda \otimes \mathcal{F}_2}\right)$$

is dense in $\mathfrak{h}_\lambda \otimes \mathcal{F}$ for almost every λ .

It remains only to pass from fibers to the direct integral. Let $\Phi \in \mathfrak{H}$ be orthogonal to $\text{Ran}(A|_{\mathfrak{H}_2})$. Testing against decomposable vectors supported on arbitrary measurable subsets of Σ shows that, for almost every λ , the fiber $\Phi(\lambda)$ is orthogonal to the dense fiber range displayed above. Therefore $\Phi(\lambda) = 0$ for almost every λ , and $\Phi = 0$. Thus $\text{Ran}(A|_{\mathfrak{H}_2})^\perp = \{0\}$, i.e. $\text{Ran}(A|_{\mathfrak{H}_2})$ is dense in $\mathfrak{H}$.

Application of Posilicano's theorem

The two density requirements in Theorem 3.3/Theorem 3.13 of [1], which restate Posilicano's abstract results [2], are now verified. Since $0 < s \leq 1$, the operator $T = -ARA^*$ is bounded exactly as in Eq. (47) of [1]. Therefore the proof of part (a) of Theorem 1.3 in [1] applies verbatim and gives self-adjointness on the stated domain.

For norm resolvent convergence, choose $f_{j,n} \in \mathcal{H}$ as in Proposition 4.1 of [1]. The estimates in Eq. (74)–(76) of [1] use only the boundedness of the finitely many B_j , not finite dimensionality. In the present setting, for $\Psi \in \mathfrak{H}_1$ and $A_n^\sharp = A_n$ or $A_n^\dagger$,

$$\|A_n^\sharp \Psi\| \leq \left(\sum_{j=1}^N \|B_j\| \|f_{j,n}\|_{\mathcal{H}} \right) \|I_{\mathfrak{h}} \otimes (\mathcal{N} + 1)^{1/2} \Psi\|,$$

and the lower bound $\omega \geq m_0 > 0$ again controls the last factor by the $\mathfrak{H}_1$ norm, up to a constant. Thus the regularized A_n are closable and infinitesimally H_{free} -bounded, while

$$\|A_n - A\|_{\mathfrak{H}_s \rightarrow \mathfrak{H}} \leq \sum_{j=1}^N \|B_j\| \|a(f_{j,n}) - a(f_j)\|_{\mathcal{F}_s \rightarrow \mathcal{F}} \rightarrow 0,$$

and similarly for the adjoints. With $E_n = 0$, the proof of part (b) of Theorem 1.3 in [1] applies unchanged, proving norm resolvent convergence.

Verification and Limitations

The proof was checked against the source proof line by line. Finite dimensionality is replaced only in Proposition 4.4, where the common eigenbasis is replaced by the joint spectral theorem. The regularization estimates already use operator norms $\|B_j\|$ and therefore survive unchanged for bounded operators on a separable spin space.

This packet does not relax normality, pairwise commutation, or the zero common kernel assumption. It also does not treat unbounded spin Hamiltonians K or the Case 2/Case 3 renormalization problems mentioned in the source.

Bounded novelty search on 2026-07-19 for exact phrases around “generalized spin-boson”, “mild ultraviolet”, “infinite-dimensional”, and “spectral calculus” found the source paper and adjacent spin-boson literature, but no later paper carrying out this precise bounded commuting-normal infinite-dimensional extension.

References

- [1] Sascha Lill and Davide Lonigro, *Self-adjointness and domain of generalized spin-boson models with mild ultraviolet divergences*, arXiv:2307.14727, 2023; version 2, 2025. <https://arxiv.org/abs/2307.14727>
- [2] Andrea Posilicano, *On the self-adjointness of $H + A^* + A$* , Mathematical Physics, Analysis and Geometry 23 (2020), no. 4, 1–31.